Table 9.2 Escape Sequences

Escape sequence	Non-graphical character
\a	Audible or alert bell
\b	Backspace
\f	Form feed
\n	Newline or linefeed
\r	Carriage return
\t	Horizontal tab
\v	Vertical tab
\\	Backslash
\'	Single quote
\"	Double quote
\?	Question Mark
\On	Octal number
\xHn	Hexadecimal number
\0	Null

Even though an escape sequence contains two characters, they should be enclosed within single quotes because, C++ consider escape sequences as character constants and allocates one byte in ASCII representation.



ASCII (American Standard Code for Information Interchange) was first developed and published in 1963 by the X3 committee, a part of the American Standards Association (ASA).

Evaluate Yourself

1. What is meant by literals? How many types of integer literals are available in C++?
2. What kind of constants are following?
   i) 26 ii) 015 iii) 0xF iv) 014.9
3. What is character constant in C++?
4. How are non graphic characters represented in C++?
5. Write the following real constants into exponent form:
   i) 32.179 ii) 8.124 iii) 0.00007
6. Write the following real constants in fractional form:
   i) 0.23E4 ii) 0.517E-3 iii) 0.5E-5
7. What is the significance of null (\0) character in a string?

String Literals

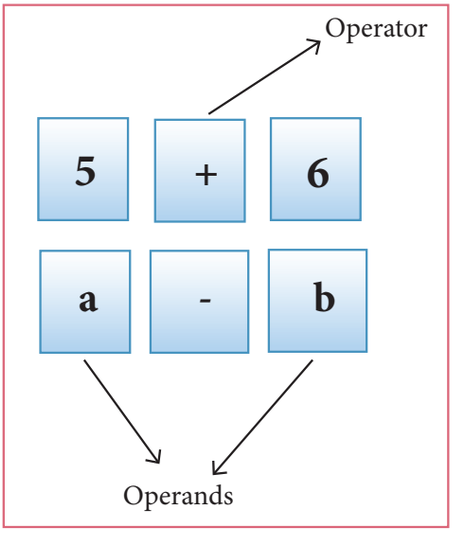
Sequence of characters enclosed within double quotes are called as String literals. By default, string literals are automatically added with a special character **'\0' (Null)** at the end. Therefore, the string "welcome" will actually be represented as "welcome\0" in memory and the size of this string is not 7 but 8 characters i.e., inclusive of the last character \0.

Valid string Literals : "A", "Welcome" "1234"

Invalid String Literals : 'Welcome', '1234'

9.3.4 Operators

The symbols which are used to do some mathematical or logical operations are called as **"Operators".** The data items or values that the oper ators act upon are called as **"Operands".**



In C++, The operators are classified on the basis of the number of operands.

(i) **Unary Operators** - Require only one operand

(ii) **Binary Operators** - Require two operands

(iii) **Ternary Operators** - Require three operands

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